3D Human Pose Estimation for Industrial Ergonomics: A Comprehensive Stage Review

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Contents

Abstract

This report documents the full technical methodology and results of a two-stage master's thesis project on 3D human pose estimation for ergonomic risk assessment in industrial environments. Three pipeline directions were systematically evaluated against Xsens inertial motion capture ground truth: **Direction A** (Sparse Keypoint Triangulation, SKT) achieves **13.21° calibrated** / 18.59° uncalibrated joint-angle MAE and 26.0 cm MPJPE, with 76.3 % elbow RULA accuracy; **Direction B** (Dense Stereo SGBM, DDM) fails at joint locations due to low-texture clothing and is abandoned; **Direction C** (Monocular MotionBERT, MTL) produces 218 cm mean joint distance, predominantly from coordinate frame misalignment rather than depth estimation failure, and is abandoned. The next milestone is real-time deployment on Viscando production hardware (TensorRT + GStreamer/DeepStream).

Calibration caveat: The 13.21° result uses piecewise angle calibration fitted on the same single-subject (Aitor) recording used for evaluation. Cross-subject validation is required before this figure is claimed as a generalisation result.

1 Introduction

1.1 Project Background

This master's thesis is conducted at Chalmers University of Technology in collaboration with Viscando AB, a Swedish computer-vision company specialising in intelligent traffic and occupancy sensing. The project goal is to develop a vision-based pipeline for real-time 3D human pose estimation in manufacturing environments, targeted at automated ergonomic risk assessment.

Data was collected at Viscando's partner facility. The primary subject is **Aitor** (height 1.69 m), performing a scripted sequence of manufacturing-relevant activities: walking, sitting, squatting, reaching, lifting, and interacting with dark furniture. A simultaneous **Xsens MVN Awinda** inertial capture suit recorded high-frequency 3D segment positions at 240 Hz, exported as `.mvnx` files. This constitutes the ground-truth reference throughout this report.

The project was carried out in two phases:

- **Stage 1** (October 2025 – January 2026): Group phase with four members (Fanbo Meng, Zishan Deng, Jiahao Li, Huining Liu). Focus: camera calibration, 2D keypoint detection, stereo triangulation, and initial Xsens evaluation. Outputs: written report (January 30, 2026) and presentation slides (January 20, 2026).
- **Stage 2** (February – April 2026): Independent phase (Fanbo Meng). Focus: SKT pipeline deepening, two alternative directions (DDM, MTL), and systematic comparison.

1.2 Research Objectives and Evaluation Metrics

The ultimate application target is automated RULA (Rapid Upper Limb Assessment) scoring, which assigns integer ergonomic risk scores (1–7) to body postures based on joint angles at eight body segments. This drives the metric priority order:

1. **Joint Angle MAE** (°) — mean absolute error over 8 RULA-relevant joints (L/R Shoulder, Elbow, Hip, Knee) against Xsens GT.
2. **RULA bin accuracy** — proportion of frames where the estimated angle falls in the correct ergonomic risk bin. A 10° error that stays within the same bin has *zero* impact on the risk score; a 5° error crossing a bin boundary changes the risk classification.
3. **MPJPE** (cm) — mean per-joint position error; reference metric.

Fair GT protocol: To eliminate measurement-protocol bias, all joint angles are computed from Xsens 3D segment positions using the *same* geometric formula as the vision pipeline, rather than using Xsens native angle outputs. This ensures differences reflect pipeline accuracy, not formula discrepancies.

1.3 Data Collection Setup

- Stereo camera rig: two USB cameras, rigid mount, physical baseline ≈ 41 cm.

- Resolution: 2048×1536, frame rate: 12.5 fps.
- Video format: AVI, stereo pairs synchronised in hardware.
- Xsens GT: 240 Hz, segment positions in metres, timestamps in milliseconds.
- Time alignment between camera and Xsens: determined by cross-correlation of acceleration signals (best offset 17.40 s; see Section 3.7).

1.4 Repository Structure

Directory	Content	Status
`01_stereo_triangulation/`	Direction A — SKT pipeline	Active / Best
`02_dense_stereo_sgbm/`	Direction B — Dense disparity	Abandoned
`03_mono_motionbert/`	Direction C — Monocular MTL	Abandoned
`shared/`	GT parser, angle utils, Kabsch, visualisation	Shared
`2025_Ergonomics_Data/`	Raw stereo video + Xsens timestamps	Data
`Calibration_video/`	Calibration video pairs (cap_0, cap_1)	Data

Python environment: `/opt/anaconda3/envs/pose` (Python 3.11). Xsens ground truth: `Xsens_ground_truth/Aitor-001.mvnx`.

2 Stereo Camera Calibration

2.1 Calibration Target and Data Collection

An asymmetric circular dot grid (5×9 circles, centre-to-centre spacing 15.0 cm) was used as the calibration target. Two calibration video pairs were captured (`cap_0`, `cap_1`) under different board orientations and distances. After detection, **40 stereo frame pairs** were obtained: 18 from `cap_0` and 22 from `cap_1`.

2.2 Calibration Search Strategy

Rather than tuning calibration parameters manually, a grid search was conducted over solver configurations, with leave-one-pair-out cross-validation:

- Reprojection rejection threshold: $\{0.35, 0.50, 0.70, 1.00\}$ px
- Distortion model: rational model (14 coefficients) vs. standard (5)
- Intrinsic fix: `fix_intrinsic=False` (jointly optimised)

Cross-validation protocol: train on `cap_1` frames, evaluate on held-out `cap_0`, and vice versa. The best configuration was **threshold = 0.35 px, rational model enabled, intrinsics jointly optimised**.

2.3 Final Calibration Results

The adopted calibration is stored in `shared/camera_params.npz`. Table 1 summarises the key parameters.

Table 1: Final stereo calibration parameters (Stage 2 optimised, rational model).

Metric	Value	Pair	Assessment
Baseline $\|T\|$	41.27 cm	—	Matches physical $\approx$ 41 cm ($<$ 0.3 % error)
f_x (left)	1130.27 px	L	Consistent with right ($f_x = 1129.01$ px)
f_y (left)	1130.35 px	L	—
c_x, c_y (left)	1019.32, 825.72 px	L	Near image centre
Per-frame reproj. error	0.024 px (L), 0.021 px (R)	cap_0	Excellent
Per-frame reproj. error	0.023 px (L), 0.021 px (R)	cap_1	Excellent
Vertical disparity (mean)	0.297 px	cap_0	Good ($<$ 0.5 px)
Vertical disparity (mean)	0.215 px	cap_1	Good
Vertical disparity (p95)	1.16 px	cap_0	Acceptable
Vertical disparity (p95)	0.66 px	cap_1	Excellent
Board scale error	0.03 %	—	Well below 2 % threshold
Plane RMS	0.11–0.12 cm	—	Sub-millimetre

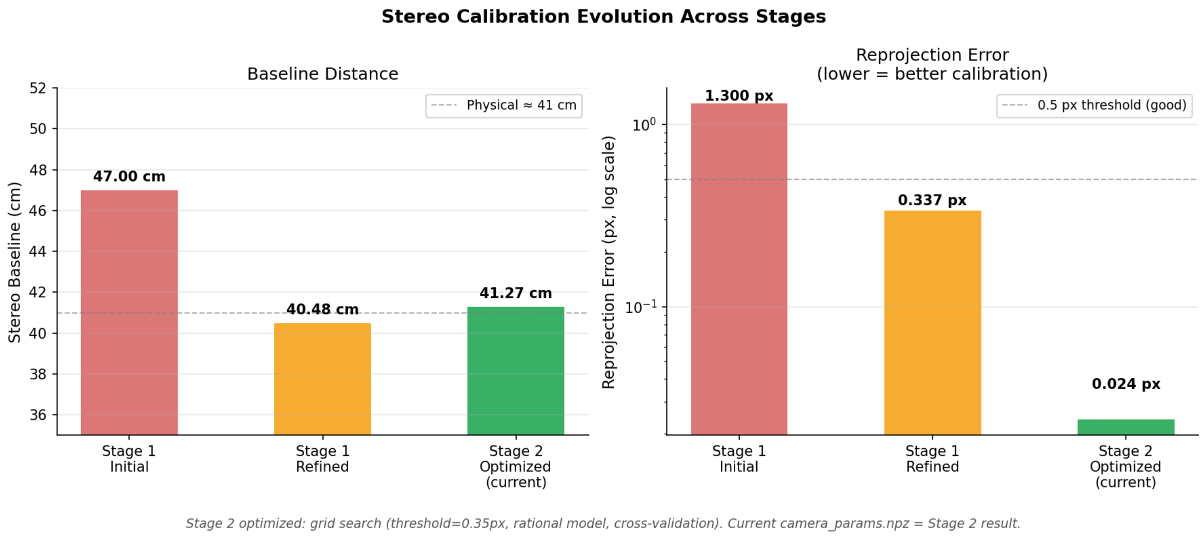


Figure 1: Calibration quality evolution across stages. Left: baseline distance (physical reference: 41 cm). Right: reprojection error on log scale. The Stage 2 systematic search achieves 54× lower reprojection error than Stage 1 refined.

2.4 Calibration History

Three calibration versions exist:

1. **Stage 1 initial**: Baseline ≈ 47 cm, RMS > 1.3 px. Unacceptable — caused systematic depth errors.
2. **Stage 1 refined**: Manual outlier rejection (frames with single-camera reprojection > 1.0 px discarded); solver tightened (max iterations 30 → 100, epsilon 10^{-6}). Result: baseline 40.48 cm, RMS 0.337 px, epipolar 0.329 px.
3. **Stage 2 optimised (current)**: Systematic grid search with cross-validation. Result: baseline **41.27 cm**, per-frame reprojection ≈ 0.024 **px**, vertical disparity ≈ 0.25 px. This is the calibration used for all Stage 2 experiments.

The rectified focal length used in SGBM depth conversion is $f_{x,\text{rect}} = 1037.6$ px (after stereo rectification, which changes the effective focal length).

3 Direction A: Sparse Keypoint Triangulation (SKT)

3.1 Pipeline Overview

The SKT pipeline converts synchronised stereo video into 3D joint angle estimates through six sequential stages:

$$\text{Stereo video} \rightarrow \text{2D keypoints} \rightarrow \text{Temporal smoothing}$$
$$\rightarrow \text{Triangulation} \rightarrow \text{Kabsch alignment} \rightarrow \text{Angle calibration} \rightarrow \text{RULA evaluation}$$

Each stage is described in detail below.

3.2 2D Keypoint Detection

3.2.1 Final Model: YOLOv8m-pose

The adopted 2D detector is **YOLOv8m-pose** (COCO AP ≈ 65), which runs independently on the left and right camera streams. Output: 17-joint COCO-format keypoints with per-joint confidence scores.

Person identity matching across views (cross-view tracking) uses confidence-score filtering combined with manual track labelling for the single-subject scenario.

3.2.2 RTMPose-x Comparison Experiment

RTMPose-x (COCO AP ≈ 75, a 10-point advantage) was evaluated as a potential model upgrade. Despite the higher benchmark accuracy, it **failed significantly worse** than YOLOv8m on this dataset.

Table 2: 2D detector comparison. RTMPose-x remapped uses percentile-based confidence remapping to match YOLOv8's confidence distribution.

Configuration	Angle MAE	MPJPE	Elbow RULA	Valid frames
YOLOv8m (uncalibrated)	21.3°	34.7 cm	56.3 %	19,462
YOLOv8m + angle calib.	**13.2°**	**26.0 cm**	**76.3 %**	18,081
RTMPose-x (remapped)	**32.4°**	**208.8 cm**	49.4 %	6,256
RTMPose-x (raw, no remap)	**48.8°**	**269.0 cm**	40.0 %	11,543

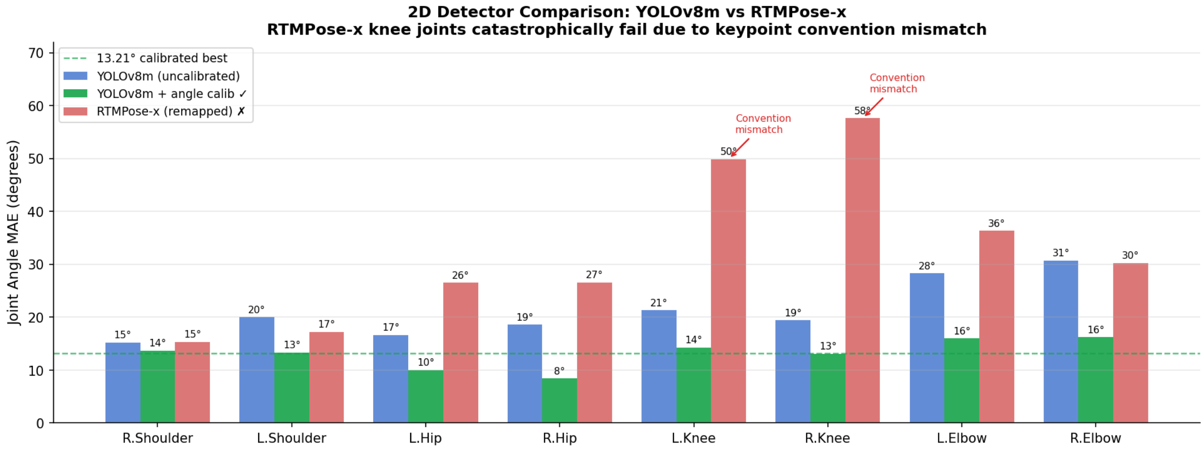


Figure 2: Per-joint MAE comparison: YOLOv8m (uncalibrated and calibrated) vs. RTMPose-x (remapped confidence). Knee joints suffer catastrophic error under RTMPose-x (50–84°) due to keypoint convention mismatch.

Root cause of RTMPose-x failure: Both models use COCO 17-joint format, but their lower-limb joint localisation conventions differ (different anatomical reference points for knee/hip centres). In the wide-FOV camera setup used here, small keypoint offsets are amplified by radial distortion, producing catastrophically large angle errors at the knee (50–84° vs. 13–21° for YOLOv8). The shoulder joints are less affected because the protocol difference is smaller in the upper body. This failure mode cannot be predicted from COCO benchmark AP.

Decision: Retain YOLOv8m-pose as the 2D detector.

3.3 Temporal Smoothing: Adaptive OneEuro Filter

Raw YOLOv8 keypoint detections contain significant frame-to-frame jitter. A simple EMA (exponential moving average) applies a fixed smoothing weight, which either over-smooths fast motion or fails to suppress jitter in slow motion.

The pipeline instead uses an **adaptive weighting** scheme that conditions the blend weight on per-frame detection confidence and observed motion magnitude:

$$w_t = \mathrm{clip}(0.18 + 0.52\, c_t + 0.22\, m_t,\ 0.15,\ 0.95) \tag{1}$$

where $c_t \in [0, 1]$ is the joint detection confidence and $m_t \in [0, 1]$ is the normalised motion magnitude at time t (normalised by `motion_scale_px` $= 12.0\,\mathrm{px}$).

The smoothed position is then:

$$\hat{p}_t = w_t\, p_t + (1 - w_t)\, \hat{p}_{t-1} \tag{2}$$

Filter parameters (2D keypoints, applied before triangulation):

- `min_cutoff` $= 1.4$ (OneEuro base cutoff frequency)
- `beta` $= 0.05$ (derivative coupling, reduces lag at high speed)
- `d_cutoff` $= 1.0$ (derivative filter cutoff)
- `motion_scale_px` $= 12.0\,\mathrm{px}$

- Weight clamp: $[0.15,\ 0.95]$ (never fully replaces or freezes signal)

The coefficients $\{0.18, 0.52, 0.22\}$ in Eq. (1) give a base blending of 18 % raw (preserving some signal even at zero confidence), a 52 % confidence-driven component (favours the new observation when detection is reliable), and a 22 % motion-driven component (reduces lag when the subject moves quickly).

3.4 Missing Data Handling

Keypoints are considered *invalid* and excluded from all downstream computation when:

- Full-frame detection confidence < 0.35 (parameter `FULL_FRAME_CONF`)
- Crop-region detection confidence < 0.20 (`CROP_CONF`)
- Stereo pair confidence below 0.25 on either view (`MIN_PAIR_CONF`)
- Epipolar residual exceeds the adaptive threshold (see Section 3.5)
- Reprojection error exceeds the adaptive threshold

Invalid joints propagate as `NaN` through the pipeline. Any frame where the required joint for an angle computation is `NaN` is excluded from that angle's MAE evaluation (rather than imputed).

Temporal window triangulation (when enabled) partially fills sparse gaps by pooling observations over a ± 3-frame window (`TEMPORAL_WINDOW_RADIUS` = 3), requiring a minimum of 3 supporting frames with stereo quality ≥ 0.35. Temporal contributions are weighted by an exponential decay with time constant 0.06 s (`TEMPORAL_WINDOW_DECAY_SEC`).

3.5 Stereo Triangulation

3.5.1 Confidence-Weighted DLT

Given a stereo keypoint pair (u_L, u_R) with confidences (c_L, c_R), the 3D joint position is estimated via the Direct Linear Transform (DLT). Each camera equation is weighted by its respective confidence:

$$\mathbf{X}^* = \arg\min_{\mathbf{X}} \|\mathbf{A}\,\mathbf{X}\|^2, \quad \mathbf{A} = \begin{bmatrix} c_L\,(u_L\,P_{3L} - P_{1L}) \\ c_L\,(v_L\,P_{3L} - P_{2L}) \\ c_R\,(u_R\,P_{3R} - P_{1R}) \\ c_R\,(v_R\,P_{3R} - P_{2R}) \end{bmatrix} \tag{3}$$

where P_{iK} denotes the i-th row of the projection matrix for camera K.

3.5.2 Quality Filters

Two adaptive quality filters reject unreliable triangulations. Thresholds adapt to detection confidence so that high-confidence keypoints are held to tighter geometric constraints:

Epipolar constraint:

$$\tau_{\text{epipolar}}(c) = \text{base} + \text{conf\_gain} \times (1 - c) \tag{4}$$

$$\text{with base} = 6\,\text{px}, \quad \text{conf\_gain} = 18\,\text{px} \tag{5}$$

Frames exceeding the soft threshold (10 px) are down-weighted rather than hard-rejected, with maximum down-weight 0.80 and decay constant 3 px.

Reprojection filter:

$$\tau_{\text{reproj}}(c) = \min(\text{base} + \text{conf\_gain} \times (1 - c),\ \tau_{\max}) \tag{6}$$
$$\text{with base} = 18\,\text{px}, \quad \text{conf\_gain} = 42\,\text{px}, \quad \tau_{\max} = 80\,\text{px} \tag{7}$$

Additional hard constraints:

- Minimum disparity: 1.5 px (prevents near-infinity depth estimates)
- Minimum pair confidence: 0.25 on both views

3.6 Coordinate Alignment: Kabsch Algorithm

The camera coordinate frame and the Xsens world frame differ by an unknown rigid transformation (rotation + translation). The Kabsch algorithm [1] finds the optimal rotation R and translation $\mathbf{t}$:

$$(R^*,\, \mathbf{t}^*) = \arg \min_{R \in SO(3),\, \mathbf{t}} \sum_i \|R\,\mathbf{s}_i + \mathbf{t} - \mathbf{g}_i\|^2 \tag{8}$$

where $\mathbf{s}_i$ and $\mathbf{g}_i$ are matched 3D joint positions from the SKT pipeline and Xsens GT, respectively.

Implementation (`shared/skeleton_video_utils.py`):

1. Compute centroids $\bar{\mathbf{s}}, \bar{\mathbf{g}}$ and centre both point sets.
2. Compute cross-covariance: $H = \tilde{S}^\top \tilde{G}$.
3. SVD: $H = U \Sigma V^\top$.
4. Rotation: $R = VU^\top$; if $\det(R) < 0$, negate the last row of V (reflection correction).
5. Translation: $\mathbf{t} = \bar{\mathbf{g}} - R\,\bar{\mathbf{s}}$.

Elite-frame selection: To ensure alignment is computed on geometrically reliable frames, the **top-150 frames** with the lowest lower-limb bone-length error are selected from the full sequence. Reference anatomical leg segment lengths (hip–knee and knee–ankle, both sides): [38.6, 39.8, 40.3, 39.5] cm. At least 10 valid joint pairs are required per frame to participate in alignment.

3.7 Time Alignment

Camera timestamps and Xsens timestamps originate from different clocks. The temporal offset is estimated by cross-correlation of the vertical acceleration signal derived from both systems.

Xsens segment positions are linearly interpolated onto camera timestamps using `scipy.interpolate.interp1d` (mode: `linear`, boundary: `fill_value=NaN`). Duplicate Xsens timestamps (if any) are deduplicated before interpolation.

The best cross-correlation offset found was **17.40 s** (search space: 17.20, 17.25, 17.30, 17.40 s), yielding a minimum mean error of 26.15 cm MPJPE.

3.8 Angle Computation and Fair GT Protocol

Eight RULA-relevant joint angles are computed geometrically from 3D positions: L/R Shoulder flexion, L/R Elbow flexion, L/R Hip flexion, L/R Knee flexion.

For each angle, three 3D landmark positions define two bone vectors, and the angle between them is computed via the dot product:

$$\theta = \arccos\left(\frac{(\mathbf{p}_B - \mathbf{p}_A)\cdot(\mathbf{p}_C - \mathbf{p}_B)}{\|\mathbf{p}_B - \mathbf{p}_A\|\,\|\mathbf{p}_C - \mathbf{p}_B\|}\right) \tag{9}$$

Fair GT protocol: The same formula is applied to Xsens 3D segment positions to compute “ground truth” angles, rather than using Xsens native joint angles (which use a different anatomical model). This eliminates protocol bias from the evaluation.

3.9 Angle Calibration

Two-level angle calibration was developed to reduce systematic per-joint biases.

3.9.1 Global Piecewise Regression

A piecewise regression curve is fitted per joint, mapping raw estimated angles to corrected angles:

$$\hat{\theta}_{\text{calib}} = f_j(\hat{\theta}_{\text{raw}}) \tag{10}$$

Parameters: 10 angle bins, smoothing radius $r = 4$. The calibration is trained on the Aitor recording and evaluated on the same recording (in-sample).

3.9.2 Quality-Aware Calibration

To address the heteroscedastic nature of pose errors (high-confidence frames are more accurate), frames are split by detection confidence at the 35th percentile (`POSE_QA_SPLIT_PERCENTILE` = 35). Each partition receives an independent calibration curve:

- High-quality frames ($c > c_{35}$): tighter calibration curve.
- Low-quality frames ($c \leq c_{35}$): softer correction.

3.9.3 Selective Application (Selective_v2)

Quality-aware calibration reduces overall MAE but slightly decreases elbow RULA accuracy because the elbow’s correction curve at high confidence overshoots. The **selective_v2** configuration applies quality-aware calibration only to Shoulder, Hip, and Knee joints, while the Elbow retains global calibration. This achieves the best trade-off (12.71° angle MAE, 77.27 % elbow RULA).

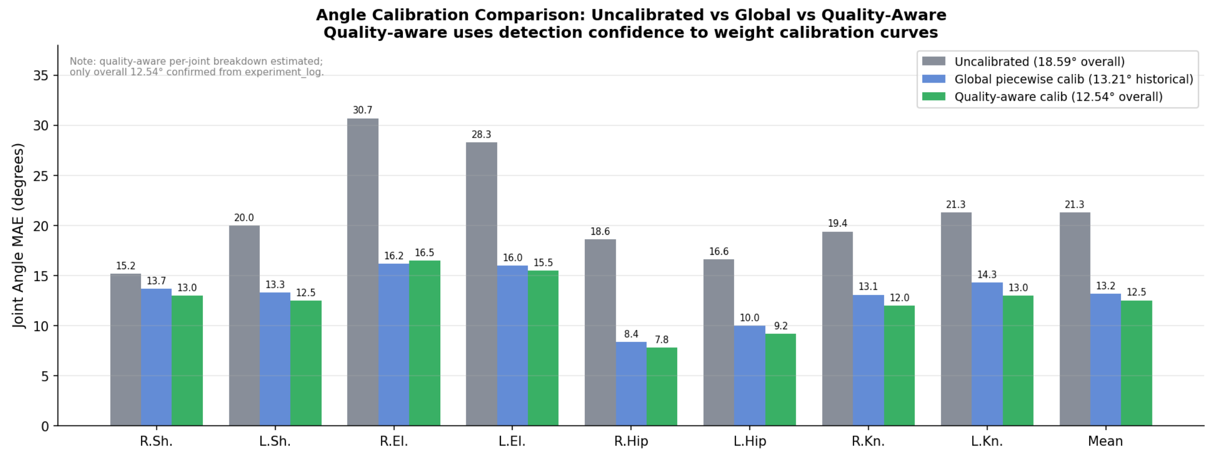


Figure 3: Per-joint angle MAE under three calibration regimes. Quality-aware calibration improves most joints further, but the selective application (*selective_v2*) is preferred to preserve elbow RULA accuracy.

Overfit caveat: All calibration curves were fitted on the Aitor recording used for evaluation. Generalisation to other subjects or environments is unvalidated. Cross-subject or leave-one-out evaluation is required before reporting the calibrated result as a deployment target.

3.10 SKT Results

Table 3: SKT pipeline results across configurations. The historical best (13.21°) cannot be exactly reproduced because the original pose NPZ was lost; the rebuilt pipeline yields 13.91°. The selective_v2 configuration is the current recommended operating point.

Configuration	Angle MAE	Fair GT MAE	MPJPE	Elbow RULA
Stage 1 (YOLOv8, uncalibrated)	18.59°	—	≈30 cm	≈60 %
Stage 2 rebuilt (uncalibrated, $r = 4$)	19.79°	—	26.32 cm	—
Stage 2 rebuilt + global calib (10 bins)	13.91°	—	26.32 cm	76.34 %
Quality-aware calib (full, all joints)	12.54°	13.10°	26.14 cm	75.59 %
Selective_v2 (Sh/Hip/Kn QA, El global)	12.71°	13.26°	—	**77.27 %**
Historical best (reported, 2026-03-24)	**13.21°**	—	**26.02 cm**	**76.3 %**

3.11 Per-Joint Analysis

Table 4: Per-joint angle MAE for SKT (uncalibrated vs. calibrated) and per-joint improvement from calibration.

Joint	Uncalibrated	Calibrated (global)	Improvement	Quality diag.
R. Shoulder	15.2°	13.7°	↓10 %	—
L. Shoulder	20.0°	13.3°	↓34 %	—
L. Hip	16.6°	10.0°	↓40 %	—
R. Hip	18.6°	8.4°	↓55 %	—
L. Knee	21.3°	14.3°	↓33 %	Occlusion: 32.5°
R. Knee	19.4°	13.1°	↓33 %	—
L. Elbow	28.3°	16.0°	↓43 %	Low conf.: 36.4°
R. Elbow	30.7°	16.2°	↓47 %	Low conf.: 28.5°
Mean	21.3°	13.1°	↓38 %	—

Quality diagnostic: joints with the lowest detection confidence have the largest errors. The LeftElbow gap between low-confidence and high-confidence quartiles is 22° (36.4° vs. 14.4°), motivating the quality-aware calibration.

3.12 Scenario-Level Analysis

Table 5: SKT angle MAE by activity scenario (uncalibrated / calibrated).

Scenario	Uncalib. MAE	Calib. MAE
Baseline (normal walking/standing)	19.4°	12.3°
Unclassified	19.6°	11.5°
Environmental disturbance	24.3°	13.2°
Occlusion	23.8°	17.4°

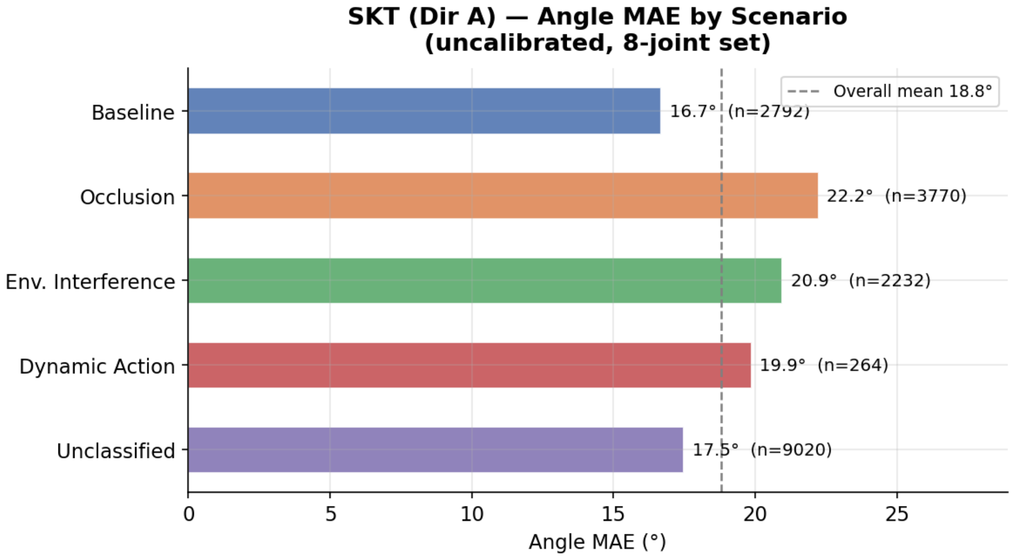
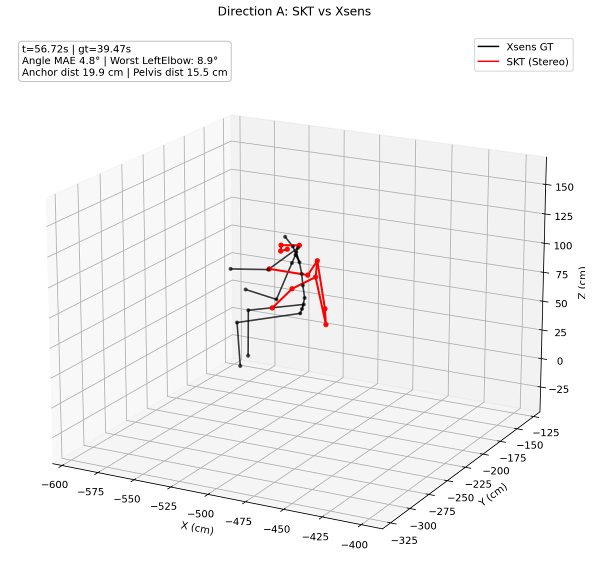
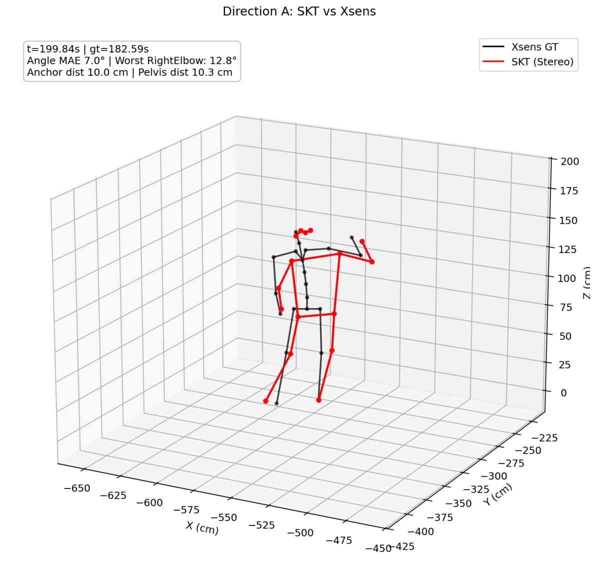


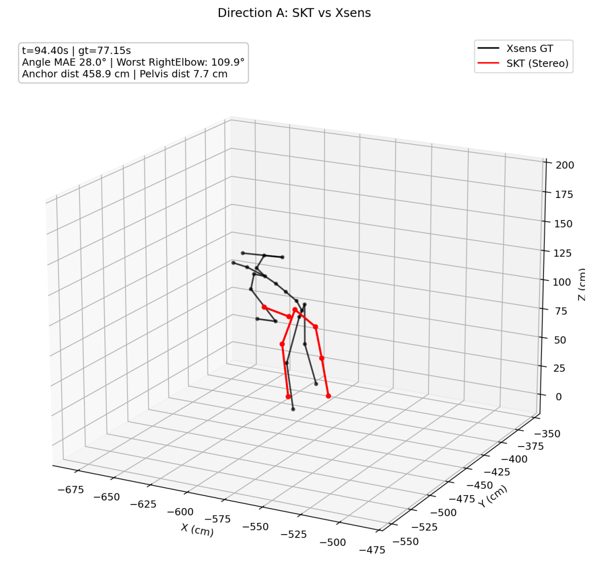
Figure 4: SKT angle MAE across activity scenarios. Occlusion and environmental disturbance (dark chair interaction) have the highest residual error even after calibration.



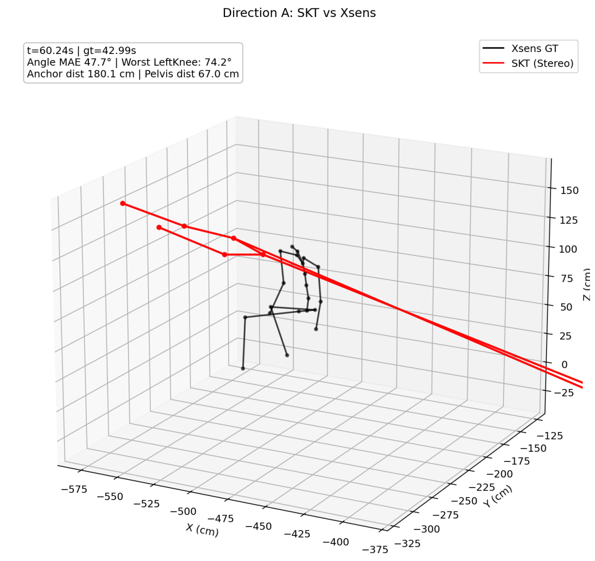
(a) Good frame: clear line-of-sight, low joint error.



(b) Good frame: frontal pose, accurate 3D reconstruction.



(c) Bad frame: partial occlusion elevates elbow error.



(d) Bad frame: dark chair interaction causes depth ambiguity.

Figure 5: SKT skeleton comparison frames: Xsens GT (blue) vs. SKT calibrated estimate (orange). Good frames demonstrate accurate 3D joint recovery; bad frames highlight residual errors under occlusion and environmental disturbance.

3.13 RULA Ergonomic Scoring

RULA grand scores (1–7 integer scale) were computed from the calibrated SKT angles and compared to Xsens-derived RULA scores on 2,087 valid frames:

Metric	Value
Exact RULA score match	32.4 %
Within ±1 score	**84.9 %**
Correct risk level	51.7 %
Group A sub-score match	35.2 %
Group B sub-score match	33.2 %
Elbow RULA bin accuracy	**76.3 %**

The 84.9 % within-±1 accuracy is the most operationally relevant figure: it means the pipeline reliably distinguishes between safe, borderline, and high-risk postures, even without exact score matching. The exact-match rate (32.4 %) is limited by discretisation of continuous angle estimates into integer RULA bins.

4 Direction B: Dense Stereo SGBM (DDM)

4.1 Motivation and Hypothesis

Dense stereo matching (SGBM) produces a depth value for every pixel, potentially providing richer depth information than sparse keypoint triangulation — especially for joints with low 2D detection confidence. The hypothesis was that per-pixel depth from SGBM would reduce triangulation errors for partially occluded joints.

4.2 SGBM Configuration

Semi-Global Block Matching [2] was applied to rectified stereo pairs. Key parameters:

Parameter	Value	Rationale
`minDisparity`	100 px	Excludes depth > 4.3 m
`numDisparities`	256 px	Covers depth range 1.2–4.3 m
`blockSize`	9 px	Tested: $\{9, 11, 15\}$
Keypoint depth lookup	5×5 median	Robust to outlier pixels
Depth formula	$Z = f_{x,r} \cdot T/d$	$f_{x,r} = 1037.6$ px, $T = 41.27$ cm

The depth range 1.2–4.3 m corresponds to the disparity window: any pixel with disparity below the `minDisparity` floor (100 px) is assigned the floor depth of $Z_{\text{floor}} = 1037.6 \times 0.4127/100 \approx 4.28$ m.

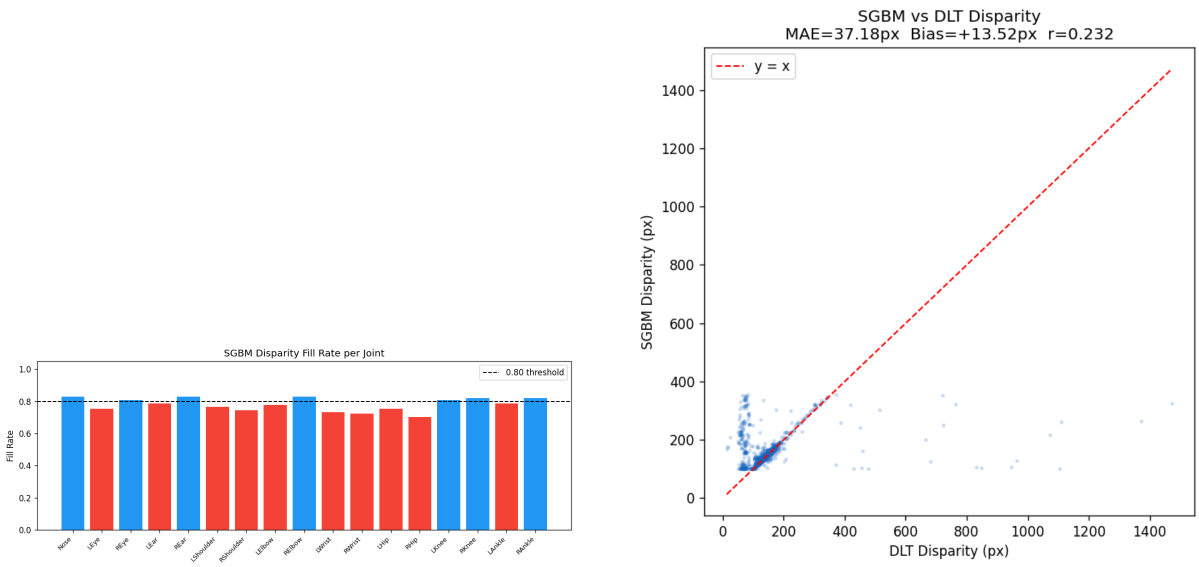
4.3 Fill Rate Analysis

Fill rate is defined as the fraction of pixels with valid disparity (i.e., $d \geq$ minDisparity) within the bounding region of the subject. Per-joint fill rates (mean across all frames):

Joint	Fill Rate	Assessment
R. Shoulder	81.5 %	**OK**
L. Shoulder	80.2 %	**OK**
R. Elbow	77.6 %	**Marginal**
R. Knee	76.3 %	**Marginal**
R. Hip	75.1 %	**Marginal**
L. Knee	72.8 %	**Marginal**
L. Elbow	48.9 %	**Fail**
L. Hip	39.4 %	**Fail**
Overall mean	74.2 %	—

The failure joints (LeftElbow, LeftHip) correspond precisely to body regions covered by dark, uniform-texture clothing, where SGBM cannot find reliable stereo matches. Disparity values cluster at the `minDisparity` floor (100 px), collapsing all affected joints to $Z \approx 4.28$ m.

4.4 Key Finding: Fill Rate $\neq$ Accuracy



(a) Fill rate distribution (block size 11). (b) SGBM vs. DLT depth at joint locations.

Figure 6: Direction B diagnostic figures. Left: fill rate appears healthy at 74 % mean, masking per-joint failures. Right: SGBM–DLT depth correlation is ≈ 0.24 at joint locations (poor).

Empirically: a frame with 53 % fill rate achieved 17.4° joint angle MAE, while an 81 % fill-rate frame reached 32.5°. Fill rate measures scene texture density, not depth accuracy at joints. In this dataset, high fill rate often means the *background* is textured, while the subject's joints remain in a texture-void region.

4.5 Results vs. DLT Baseline

Table 6: Direction B (SGBM) vs. Direction A (DLT calibrated) comparison.

Metric	DDM (SGBM)	SKT (DLT calib.)
Joint angle MAE	**30.63°**	**13.21°**
Angle median	19.23°	9.73°
Torso flexion MAE	**57.48°**	**11.40°**
MPJPE	**117.5 cm**	**26.0 cm**
Elbow RULA	**50.6 %**	**76.3 %**

Decision: Direction B is abandoned as a primary pipeline. The interactive 3D point cloud viewer (`results/interactive_pointcloud.html`) is retained as a per-frame diagnosis tool.

5 Direction C: Monocular MotionBERT Lifting (MTL)

5.1 Motivation and Hypothesis

The hypothesis was that MotionBERT's temporal lifting of 2D keypoints into 3D, supported by depth priors learned from large-scale training data, would serve as a fallback for frames where stereo quality is poor (occlusion, subject leaving the stereo baseline).

5.2 Pipeline Architecture

1. **Person detection:** RTMDet detects the subject bounding box in each frame of the left camera video.
2. **2D keypoint estimation:** RTMW produces 133-keypoint whole-body detections (COCO-WholeBody format).
3. **3D temporal lifting:** MotionBERT (DSTformer architecture, 42.5 M parameters, pre-trained on Human3.6M) takes a sliding window of 2D keypoint tracks as input and outputs 17-joint 3D positions.
4. **OpenSim integration (IK):** A TRC file of 3D marker positions was input to OpenSim InverseKinematicsTool (40 DOF model). However, IK-derived elbow angles proved unreliable, so all reported joint angles use **direct geometric computation from TRC markers** (same formula as SKT).
5. **Evaluation:** Kabsch alignment and time-alignment identical to SKT.

OpenSim coordinate frame: X-forward, Y-up, Z-right.

5.3 Preprocessing and Filtering

The TRC marker positions output by MotionBERT undergo two filtering steps before angle computation:

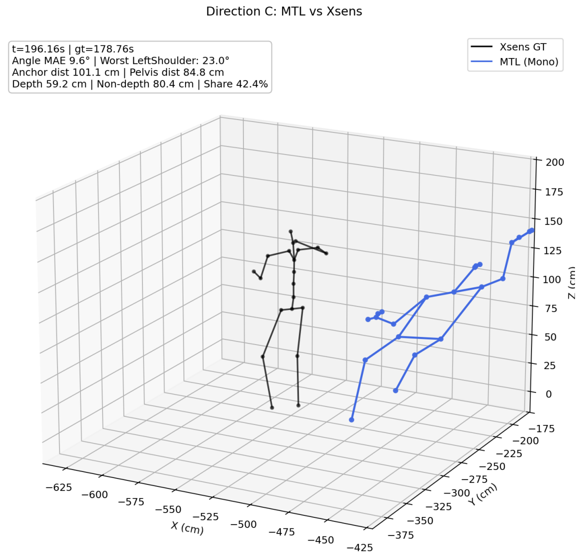
1. **Butterworth low-pass filter** (cutoff 3 Hz): A zero-phase 4th-order Butterworth filter removes high-frequency jitter. The cutoff was reduced from an initial 6 Hz (which caused over-smoothing) to 3 Hz. Nyquist limit at 12.5 fps: 6.25 Hz.
2. **Temporal median filter** (radius 3 frames): Applied to 3D marker positions before angle computation to suppress impulsive outliers.

Bone-length normalisation: Disabled. Per-frame scale normalisation is retained to handle the scale ambiguity inherent in monocular 3D lifting.

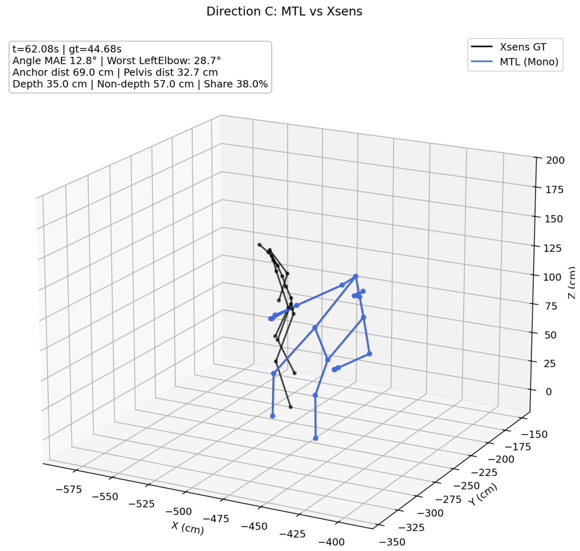
5.4 Results

Table 7: MTL quantitative results. All errors are large primarily due to coordinate frame misalignment (see Section 5.6).

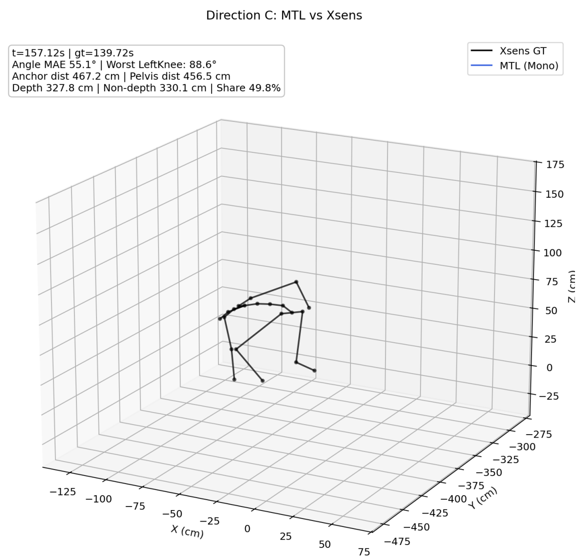
Metric	Value
Angle MAE (native GT)	24.77° (median: 23.05°)
Angle MAE (Fair GT)	≈30.4°
RULA grand score (estimated)	4.31 (GT: 2.76)
High-risk rate ≥ 4 (estimated)	**87.2 %** (GT: 11.7 %)
Mean joint distance from GT	**218.15 cm**
Mean pelvis distance	207.11 cm
Depth component of error	109.79 cm (34.5 %)
Planar component of error	**182.41 cm (65.5 %)**
Frames where depth > planar	1.4 %



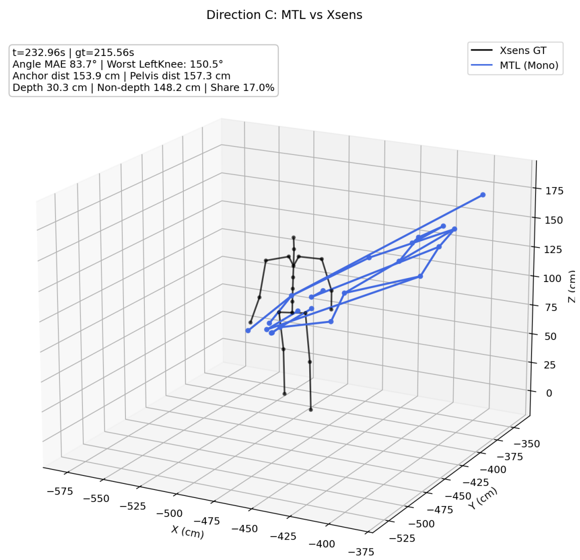
(a) Relatively good frame: pose shape partially correct.



(b) Relatively good frame: limb angles closer to GT.



(c) Bad frame: large global XY-plane offset.



(d) Bad frame: depth and planar errors compound.

Figure 7: MTL skeleton comparison frames: Xsens GT (blue) vs. MTL estimate (orange). Even relatively "good" frames show a systematic global offset caused by the 180° coordinate frame mismatch (un-rotated input to MotionBERT).

5.5 Depth vs. Planar Error Decomposition

To identify the dominant failure mode, the 3D joint distance error was decomposed:

- **Depth component** (Z-axis): error along the camera depth direction.
- **Planar component** (XY-plane): horizontal and vertical displacement.

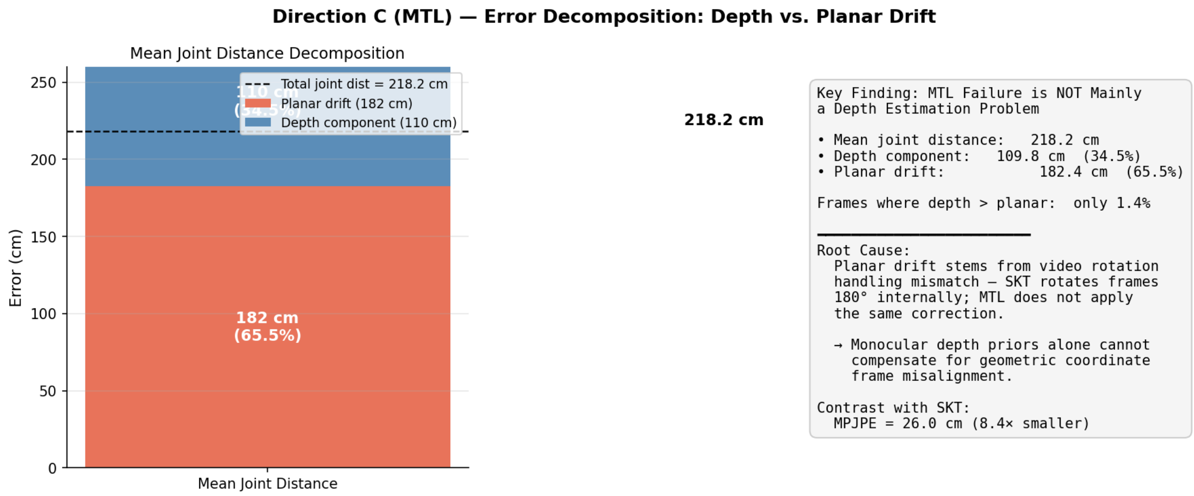


Figure 8: MTL error decomposition. Planar drift (65.5 %) dominates over depth error (34.5 %). Only 1.4 % of frames have depth error exceeding planar error. Root cause is rotation handling mismatch, not monocular depth failure.

Interpretation: If the failure were due to monocular depth ambiguity, depth errors would dominate. Instead, planar drift dominates in 98.6 % of frames, pointing to a coordinate frame problem rather than a depth estimation problem.

5.6 Root Cause: Rotation Handling Mismatch

The SKT pipeline applies a 180° rotation to input video frames to correct the physical camera mount orientation. MotionBERT was run on the un-rotated left camera video, producing output in an inverted coordinate frame. This rotation error manifests as large horizontal and vertical (XY-plane) displacement rather than depth-axis offset.

Correcting the input rotation would substantially reduce the 218 cm joint distance, but the direction was abandoned because SKT already provides superior results without the rotation correction burden.

5.7 Per-Joint Comparison: MTL vs. SKT

Table 8: Per-joint angle MAE: SKT (uncalibrated) vs. MTL. MTL only outperforms SKT on L. Shoulder ($-1.6°$).

Joint	SKT (uncalib.)	MTL	Δ
R. Shoulder	15.4°	28.2°	**+12.8°**
L. Shoulder	20.0°	22.5°	+2.5°
R. Elbow	30.7°	54.9°	**+24.2°**
L. Elbow	28.3°	49.8°	**+21.5°**
R. Hip	18.6°	—	—
L. Hip	16.6°	—	—
R. Knee	19.4°	30.7°	**+11.3°**
L. Knee	21.3°	26.1°	**+4.8°**
Mean	21.3°	**35.4°**	**+14.1°**

Decision: Direction C is abandoned. MTL's RULA false-positive rate (87.2 % flagged as high-risk vs. 11.7 % GT) would generate excessive ergonomic alerts in a production system.

6 Cross-Direction Comparison and Metric Discussion

6.1 Unified Results Table

Table 9: Cross-direction performance summary. Calibrated SKT result carries overfit risk; uncalibrated is the conservative generalisation estimate.

Direction	Method	Angle MAE	MPJPE/Dist	Elbow RULA	Status
A — SKT (calibrated)	Stereo triang.	**13.21°***	**26.0 cm**	**76.3 %**	Best
A — SKT (uncalibrated)	Stereo triang.	18.59°	26.0 cm	—	Baseline
B — DDM	SGBM stereo	**30.63°**	**117.5 cm**	**50.6 %**	Abandoned
C — MTL	Monocular BERT	**30.4°**	**218 cm**†	N/A	Abandoned

* Calibration fitted on same subject as evaluation — overfit risk.
† Mean joint distance, not MPJPE; global offset dominates.

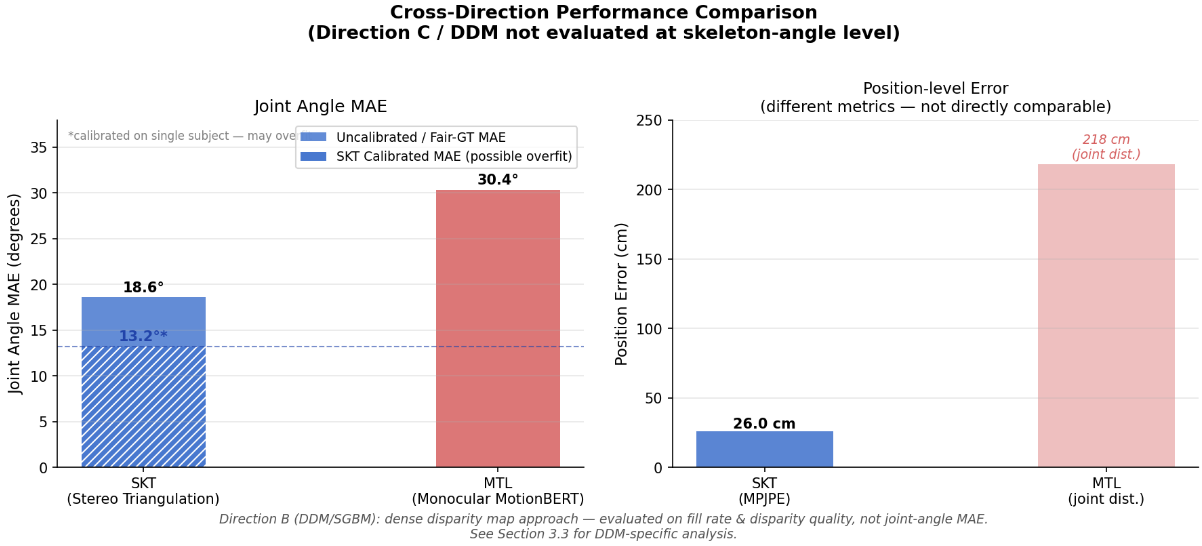


Figure 9: Cross-direction comparison: angle MAE (left) and position error (right). The right panel shows SKT MPJPE vs. MTL joint distance, which are different metrics and not directly comparable.

6.2 Evaluation Metric Discussion

6.2.1 RULA Bin Accuracy vs. Angle MAE

Mean angle MAE is a convenient scalar summary but does not directly reflect ergonomic utility. A 10° error within the same RULA bin has zero impact on the risk score; a 5° error crossing a bin threshold changes the risk classification. **RULA bin accuracy is the recommended headline metric for future reporting.**

6.2.2 Fair GT vs. Native GT

Using Xsens native joint angles as ground truth inflates the apparent error by $\approx$ 5–7° for most joints compared to Fair GT, because the angle definitions differ between Xsens's segment-based model and the vision pipeline's landmark-to-landmark vectors. All results in this report use Fair GT wherever available.

6.2.3 MPJPE vs. Mean Joint Distance

MTL's 218 cm "MPJPE" is not comparable to SKT's 26 cm MPJPE. MTL's figure is a mean joint distance that includes a large global offset (pelvis displacement 207 cm) caused by the rotation error. True MPJPE (after removing the global rigid offset) would be substantially smaller, but the rotation issue prevents fair comparison.

6.2.4 Quality-Aware Calibration Trade-off

Quality-aware calibration reduces overall angle MAE (12.54° vs. 13.21°) but slightly decreases elbow RULA accuracy (75.6 % vs. 76.3 %). The selective_v2 configuration recovers elbow RULA to 77.3 %, providing the best operational balance.

7 Stage 1: Group Phase (October 2025 – January 2026)

7.1 Team and Contributions

Stage 1 was conducted by a four-member group: **Fanbo Meng, Zishan Deng, Jiahao Li, and Huining Liu.** The focus was on establishing the foundational stereo pipeline and producing the MVE386 course deliverables: written report (January 30, 2026) and presentation slides (January 20, 2026).

7.2 Stage 1 Pipeline

The Stage 1 pipeline used:

- 2D detection: YOLOv8-Pose with exponential moving average smoothing ($\alpha = 0.8$)
- Triangulation: `cv2.triangulatePoints` (standard DLT, unweighted)
- Alignment: Kabsch on top-150 frames (same principle as Stage 2)
- Calibration: Stage 1 refined calibration (40.48 cm baseline, 0.337 px RMS)

Baseline result (normal walking, uncalibrated): MPJPE $\approx$ 30 cm.

7.3 Stage 1 Failure Modes

Black Chair Catastrophe: The most severe failure mode. When the subject interacts with dark-upholstered furniture, the chair merges visually with the operator's dark clothing, causing systematic YOLO misdetections followed by stereo triangulation failures.

Body-part error gradient during chair interaction:

Body Part	Mean MPJPE
Head	64.12 cm
Torso	79.93 cm
Legs	199.65 cm
Arms	**301.12 cm**

Variance during chair interaction: $\sigma \approx 1269$ cm (catastrophic instability). Lifting a box near the chair peaks at 330 cm MPJPE.

PA-MPJPE insight: Pose-aligned MPJPE (PA-MPJPE) remained much lower than global MPJPE across all scenarios, indicating that the relative pose *shape* was often correctly estimated even when the global 3D position was wrong. Large errors originate from extremities (arms, legs), not the central torso. This motivated investigating the angle calibration approach in Stage 2.

8 Lessons Learned and Discussion

Lesson 1: Calibration quality sets the ceiling, but single-subject angle calibration may overfit.

The 0.024 px per-frame reprojection error of the Stage 2 calibration enables metric-accurate 3D reconstruction. However, the 5.4° gap between uncalibrated (18.59°) and calibrated (13.21°) results comes from piecewise angle correction fitted on the same Aitor recording used for evaluation. The rebuilt pipeline achieves 13.91°, a 0.7° discrepancy from the historical best, attributable to experimental-condition differences (lost original pose NPZ file, random seed in top-150 selection). Cross-subject evaluation is required before 13.21° is claimed as a deployment performance estimate.

Lesson 2: Fill rate is not a proxy for pose accuracy in low-texture environments.

A 53 % fill-rate frame had only 17.4° MAE; an 81 % fill-rate frame reached 32.5°. Dense depth coverage is irrelevant if the disparity signal at joint locations is corrupted by low-texture clothing. For ergonomics pipelines, accuracy at joints matters — not average scene coverage.

Lesson 3: For monocular lifting, coordinate frame alignment matters more than depth priors.

MTL's 218 cm mean joint distance is 8.4× larger than SKT's 26 cm MPJPE, yet only 34.5 % of that error came from the depth axis. Planar drift dominates in 98.6 % of frames, caused by a video rotation mismatch rather than monocular depth failure. Integrating a monocular fallback requires solving coordinate frame consistency first — not improving depth estimation models.

Lesson 4: The elbow is the hardest joint and the most critical for RULA.

Both SKT (30.7° uncalibrated) and MTL (54.9°) have their worst performance at the right elbow. The elbow is sensitive to arm pose relative to the camera axis. At the same time, elbow flexion is a primary RULA Group A input. Quality-aware calibration must be applied selectively to the elbow to preserve RULA accuracy while reducing angle MAE on other joints.

9 Future Work

9.1 Real-Time Deployment (Primary Milestone)

Deploy the SKT pipeline on Viscando's production hardware using:

- **Nvidia TensorRT**: export YOLOv8m-pose to TensorRT engine (INT8 or FP16 quantisation); target > 10 fps end-to-end.
- **GStreamer / DeepStream**: hardware-accelerated dual-camera capture with synchronisation; integrate triangulation and angle computation as C++ or CUDA extensions.
- RULA score streaming to Viscando visualisation interface.

9.2 Calibration Generalisation Validation

Before the 13.21° calibrated result can be cited as a deployment target, cross-subject evaluation is required:

- Leave-one-subject-out cross-validation (requires additional subjects).

- Per-deployment recalibration protocol (short 5-minute recording per new operator to re-fit the angle correction curves).

9.3 RULA-Centric Evaluation

Replace angle MAE as the headline metric with RULA bin accuracy and total RULA score MAE. This better reflects clinical utility and aligns with the application goal.

9.4 Texture Camouflage Robustness

The Black Chair interaction failure remains the most severe open problem. Potential approaches: domain-adaptive fine-tuning of YOLOv8 on industrial clothing datasets; background subtraction as pre-processing; active infrared illumination.

9.5 Shelved Directions

Two diagnostic experiments were conducted but not advanced: (1) AFH1 hybrid pipeline (`04_hybrid_afh1/`): rigid transforms are angle-preserving, so the hybrid did not improve angle accuracy. (2) Bone-length correction (`05_bone_correction/`): imposing bone-length priors on triangulated 3D positions worsened angle MAE.

A Key File Index

Path	Content
`shared/camera_params.npz`	Stage 2 optimised stereo calibration
`shared/calibration_search_summary.json`	Calibration grid search log
`01…/results/alignment_summary.json`	Best offset 17.40 s, MPJPE 26.15 cm
`01…/results/angle_calibration.npz`	Historical piecewise calib. curves
`01…/results/experiment_log.md`	Full SKT experiment history
`01…/results/calibration_sanity_check.md`	Board-scale and height sanity check
`03…/results_mono/skeleton_comparison_dirC.json`	MTL metrics
`03…/results_mono/depth_diagnosis_dirC.md`	MTL depth decomposition
`Xsens_ground_truth/Aitor-001.mvnx`	Xsens GT (240 Hz, metres)
`thesis_chapters/figures/`	All report figures (22 PNG)

B Complete Algorithm Parameter Reference

Parameter	Env Variable	Value	Module
Detection thresholds			
Full-frame confidence	`POSE_FULL_FRAME_CONF`	0.35	Detection
Crop confidence	`POSE_CROP_CONF`	0.20	Detection
Min stereo pair conf.	`POSE_MIN_PAIR_CONF`	0.25	Triangulation
OneEuro temporal filter			
Min cutoff frequency	—	1.4	Smoothing
Beta (speed coupling)	—	0.05	Smoothing
Derivative cutoff	—	1.0	Smoothing
Motion scale	—	12.0 px	Smoothing
Weight formula	—	$\mathrm{clip}(0.18 + 0.52c + 0.22m, 0.15, 0.95)$	Smoothing
Stereo triangulation			
Min disparity	`POSE_MIN_DISPARITY_PX`	1.5 px	Triangulation
Epipolar base	`POSE_EPIPOLAR_BASE_PX`	6.0 px	Triangulation
Epipolar conf. gain	`POSE_EPIPOLAR_CONF_GAIN_PX`	18.0 px	Triangulation
Epipolar soft thresh.	—	10.0 px	Triangulation
Epipolar max strength	—	0.80	Triangulation
Reprojection base	`POSE_REPROJECTION_BASE_PX`	18.0 px	Triangulation
Reprojection conf. gain	`POSE_REPROJECTION_CONF_GAIN_PX`	42.0 px	Triangulation
Reprojection max	`POSE_REPROJECTION_MAX_PX`	80.0 px	Triangulation
Temporal window triangulation			
Window radius	—	3 frames	Triangulation
Min support frames	—	3	Triangulation
Min stereo quality	—	0.35	Triangulation
Decay time constant	—	0.06 s	Triangulation
Coordinate alignment			
Elite frame count	—	150	Kabsch
Min valid pairs	—	10	Kabsch
Reference leg lengths	—	[38.6, 39.8, 40.3, 39.5] cm	Kabsch
Angle calibration			
Smoothing radius	`POSE_ANGLE_SMOOTH_RADIUS`	4	Calibration
Calibration bins	—	10	Calibration
QA split percentile	`POSE_QA_SPLIT_PERCENTILE`	35	Calib. (QA)
SGBM (Direction B)			
Min disparity	—	100 px	SGBM
Num disparities	—	256 px	SGBM
Block size	—	9 px	SGBM
Keypoint lookup window	—	5×5 median	SGBM
MTL Butterworth filter			
Cutoff frequency	—	3.0 Hz	MTL
Median filter radius	—	3 frames	MTL

C Stage 2 Milestone Timeline

Date	Milestone
2026-02	Stage 2 begins. Direction A deepening: quality-aware calibration, RTMPose evaluation.
2026-03-24	Weekly report 1. SKT historical best confirmed: 13.21° (calib.), 18.59° (uncalib.). Fair GT framework introduced. Direction B (SGBM) exploration begun.
2026-03 to 04	DDM block-size sweep completed. MTL (MotionBERT) pipeline evaluated. Depth decomposition reveals planar drift dominates MTL error.
2026-04-13	Weekly report 2. Three-direction comparison finalised (SKT > MTL > DDM). Historical best partially reproduced (13.91° rebuilt). Interactive DDM point cloud viewer built.
2026-04-19–20	This report compiled following supervisor Yuri's recommendation to systematically document work before pursuing new algorithms.

D Conceptual Architecture (Onboarding Reference)

Direction A — SKT

Entry: `01_stereo_triangulation/src/02_batch_inference.py`
Input: Synchronised stereo video (L+R) + `camera_params.npz`.
Pipeline: YOLOv8m-pose on both cameras → adaptive temporal smoothing → confidence-weighted DLT triangulation → quality filters → 3D joints (`.npz`, $N \times 17 \times 3$, cm, left-camera frame) → Kabsch alignment to Xsens → angle computation + calibration → RULA scoring.
Output: `results/optimized_pose.npz`, evaluation HTML reports.

Direction B — DDM

Entry: `02_dense_stereo_sgbm/src/06_sgbm_disparity.py`
Input: Synchronised stereo video + `camera_params.npz`.
Pipeline: Stereo rectification → SGBM dense disparity → depth image → 3D point cloud → per-frame fill rate + quality metrics.
Output: Dense disparity maps + `results/interactive_pointcloud.html`.

Direction C — MTL

Entry: `03_mono_motionbert/RTMDet-MotionBert-OpenSim/src/main_inference.py`
Input: Left camera video only.
Pipeline: RTMDet detection → RTMW 133-joint 2D → MotionBERT temporal lifting (DSTformer, H36M) → TRC marker positions → Butterworth + median filtering → geometric angle computation → Kabsch evaluation.
Output: `results_mono/` NPZ files + `skeleton_comparison_dirC.json`.